

Day 4 and Day 5 Hacker Rank

Python has built-in string validation methods for basic data. It can check if a string is composed of alphabetical characters, alphanumeric characters, digits, etc.

[str.isalnum\(\)](#)

This method checks if all the characters of a string are alphanumeric (*a-z, A-Z and 0-9*).

```
>>> print 'ab123'.isalnum()
```

True

```
>>> print 'ab123#'.isalnum()
```

False

[str.isalpha\(\)](#)

This method checks if all the characters of a string are alphabetical (*a-z and A-Z*).

```
>>> print 'abcD'.isalpha()
```

True

```
>>> print 'abcd1'.isalpha()
```

False

[str.isdigit\(\)](#)

This method checks if all the characters of a string are digits (*0-9*).

```
>>> print '1234'.isdigit()
```

True

```
>>> print '123edsd'.isdigit()
```

False

[str.islower\(\)](#)

This method checks if all the characters of a string are lowercase characters (*a-z*).

```
>>> print 'abcd123#'.islower()
```

True

```
>>> print 'Abcd123#'.islower()
```

False

[str.isupper\(\)](#)

This method checks if all the characters of a string are uppercase characters (A-Z).

```
>>> print 'ABCD123#'.isupper()
```

True

```
>>> print 'Abcd123#'.isupper()
```

False

Task 1

You are given a string s .

Your task is to find out if the string s contains: *alphanumeric characters, alphabetical characters, digits, lowercase and uppercase characters*.

Input Format

A single line containing a string .

Constraints

$$0 < \text{len}(S) < 1000$$

Output Format

In the first line, print True if `s` has any *alphanumeric characters*. Otherwise, print False.

In the second line, print True if `s` has any *alphabetical characters*. Otherwise, print False.

In the third line, print True if `s` has any *digits*. Otherwise, print False.

In the fourth line, print True if `s` has any *lowercase characters*. Otherwise, print False.

In the fifth line, print True if `s` has any *uppercase characters*. Otherwise, print False.

Sample Input

qA2

Sample Output

True

True

True

True

True

HackerRank | Prepare | Python | Strings | String Validators | Exit Full Screen View

Problem

```
>>> print 'ABCD123#'.isupper()
True
>>> print 'Abcd123#'.isupper()
False
```

Task

You are given a string `S`.

Your task is to find out if the string `S` contains: alphanumeric characters, alphabetical characters, digits, lowercase and uppercase characters.

Input Format

A single line containing a string `S`.

Constraints

$0 < \text{len}(S) < 1000$

Output Format

In the first line, print True if `S` has any alphanumeric characters. Otherwise, print False.

In the second line, print True if `S` has any alphabetical characters. Otherwise, print False.

In the third line, print True if `S` has any digits. Otherwise, print False.

In the fourth line, print True if `S` has any lowercase characters. Otherwise, print False.

In the fifth line, print True if `S` has any uppercase characters. Otherwise, print False.

Sample Input

```
qA2
```

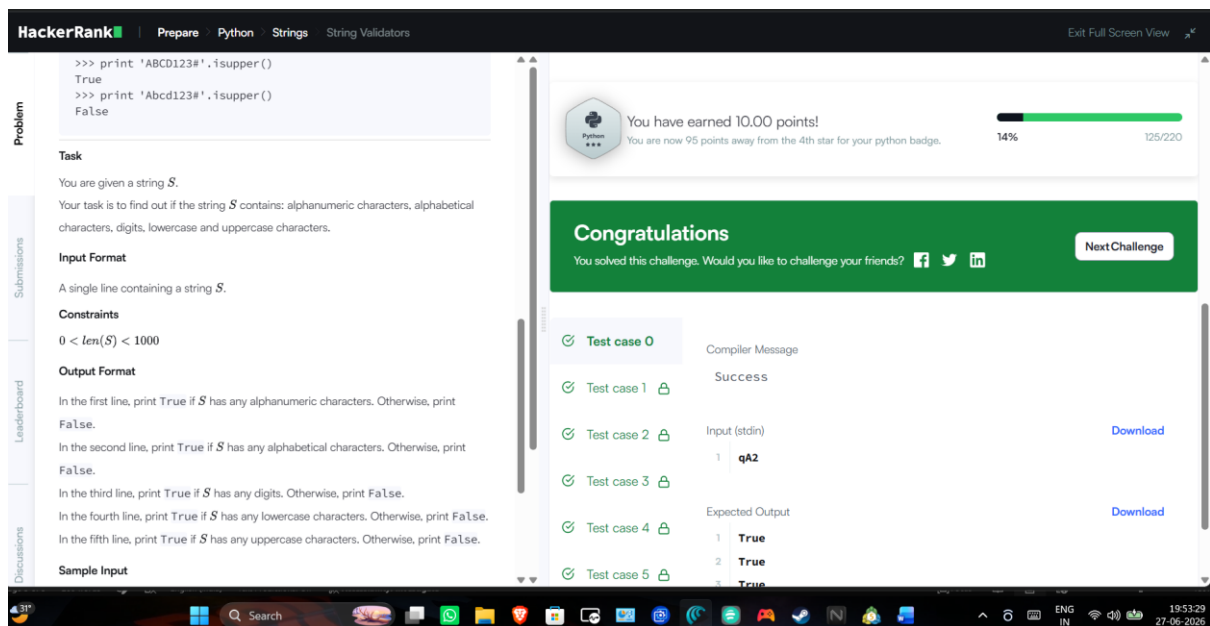
Code Editor

```
1 if __name__ == '__main__':
2     s = input()
3     isalnum = False
4     isalpha = False
5     isdigit = False
6     islower = False
7     isupper = False
8     for c in s:
9         if c.isalnum():
10            isalnum = True
11        if c.isalpha():
12            isalpha = True
13        if c.isdigit():
14            isdigit = True
15        if c.islower():
16            islower = True
17        if c.isupper():
18            isupper = True
19
20 print (isalnum)
21 print (isalpha)
22 print (isdigit)
23 print (islower)
24 print (isupper)
```

Line: 20 Col: 16

Upload Code as File | Test against custom input | Run Code | Submit Code

Windows taskbar: 19:53:06 27-06-2026



Task 2

You are given a string and your task is to *swap cases*. In other words, convert all lowercase letters to uppercase letters and vice versa.

For Example:

Www.HackerRank.com → wWw.hAcKerRaNk.CoM

Pythonist 2 → pYTHONIST 2

Function Description

Complete the *swap_case* function in the editor below.

swap_case has the following parameters:

- *string s*: the string to modify

Returns

- *string*: the modified string

Input Format

A single line containing a string .

Constraints

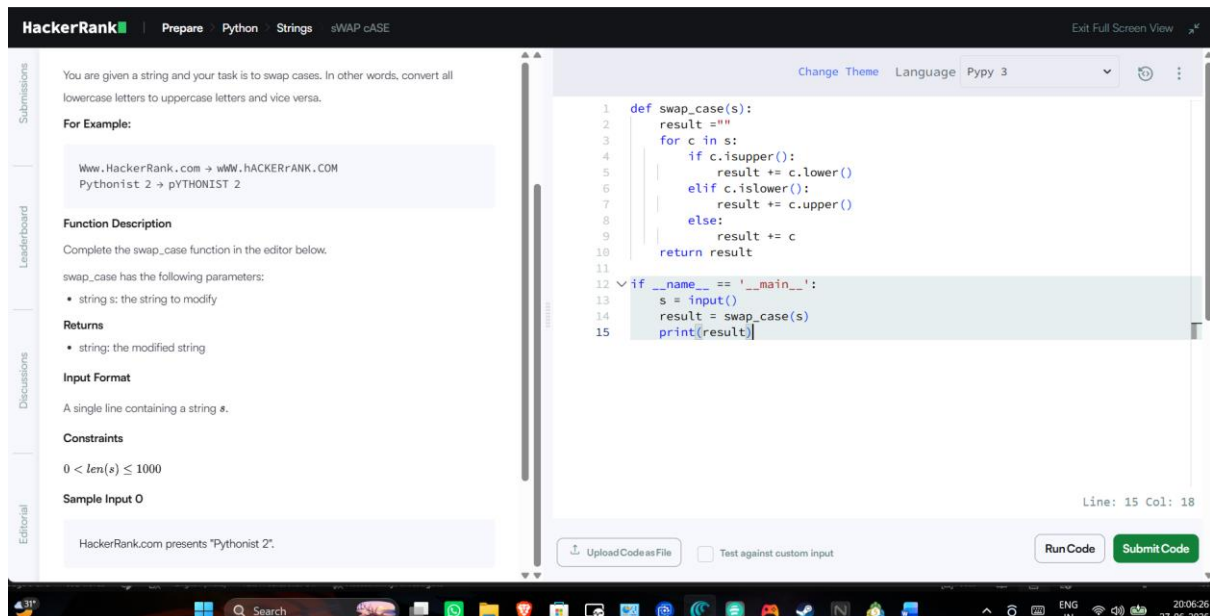
$$0 < \text{len}(s) \leq 1000$$

Sample Input 0

HackerRank.com presents "Pythonist 2".

Sample Output 0

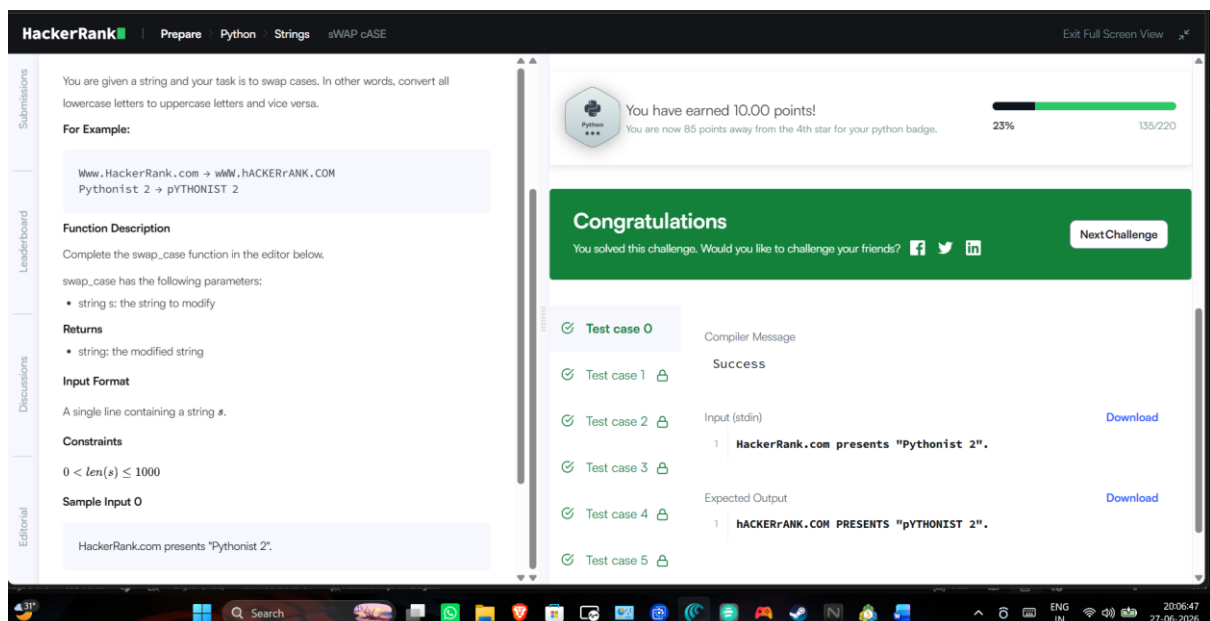
hACKERrANK.COM PRESENTS "pYTHONIST 2".



The screenshot shows the HackerRank editor interface for the 'swap_case' challenge. On the left, the problem description is visible, including the example: 'www.HackerRank.com → wWw.hACKERrANK.COM' and 'Pythonist 2 → pYTHONIST 2'. The main editor area contains the following Python code:

```
1 def swap_case(s):
2     result = ""
3     for c in s:
4         if c.isupper():
5             result += c.lower()
6         elif c.islower():
7             result += c.upper()
8         else:
9             result += c
10    return result
11
12 if __name__ == '__main__':
13     s = input()
14     result = swap_case(s)
15     print(result)
```

The code is written in Python 3. The bottom of the editor shows a status bar with 'Line: 15 Col: 18' and buttons for 'Upload Code as File', 'Test against custom input', 'Run Code', and 'Submit Code'.



The screenshot shows the HackerRank interface after submitting the solution. A notification at the top states: 'You have earned 10.00 points! You are now 85 points away from the 4th star for your python badge.' A progress bar shows 23% completion (135/220 points). A green banner reads 'Congratulations You solved this challenge. Would you like to challenge your friends?' with social media icons and a 'Next Challenge' button. Below this, the test results are displayed:

Test Case	Status	Compiler Message	Input (stdin)	Expected Output
Test case 0	Success	Success		
Test case 1	Success			
Test case 2	Success			
Test case 3	Success		HackerRank.com presents "Pythonist 2".	hACKERrANK.COM PRESENTS "pYTHONIST 2".
Test case 4	Success			
Test case 5	Success			

Each test case has a 'Download' link next to it. The bottom of the interface shows the Windows taskbar with the time 20:06:47 on 27-06-2026.

Task3

You are given a string. Split the string on a " " (space) delimiter and join using a - hyphen.

Function Description

Complete the *split_and_join* function in the editor below.

`split_and_join` has the following parameters:

- *string line*: a string of space-separated words

Returns

- *string*: the resulting string

Input Format

The one line contains a string consisting of space separated words.

Sample Input

this is a string

Sample Output

this-is-a-string

HackerRank | Prepare | Python | Strings | String Split and Join

Submissions | Leaderboard | Discussions | Editorial

Task
You are given a string. Split the string on a " " (space) delimiter and join using a = hyphen.

Function Description
Complete the `split_and_join` function in the editor below.
`split_and_join` has the following parameters:

- *string line*: a string of space-separated words

Returns
• *string*: the resulting string

Input Format
The one line contains a string consisting of space separated words.

Sample Input
this is a string

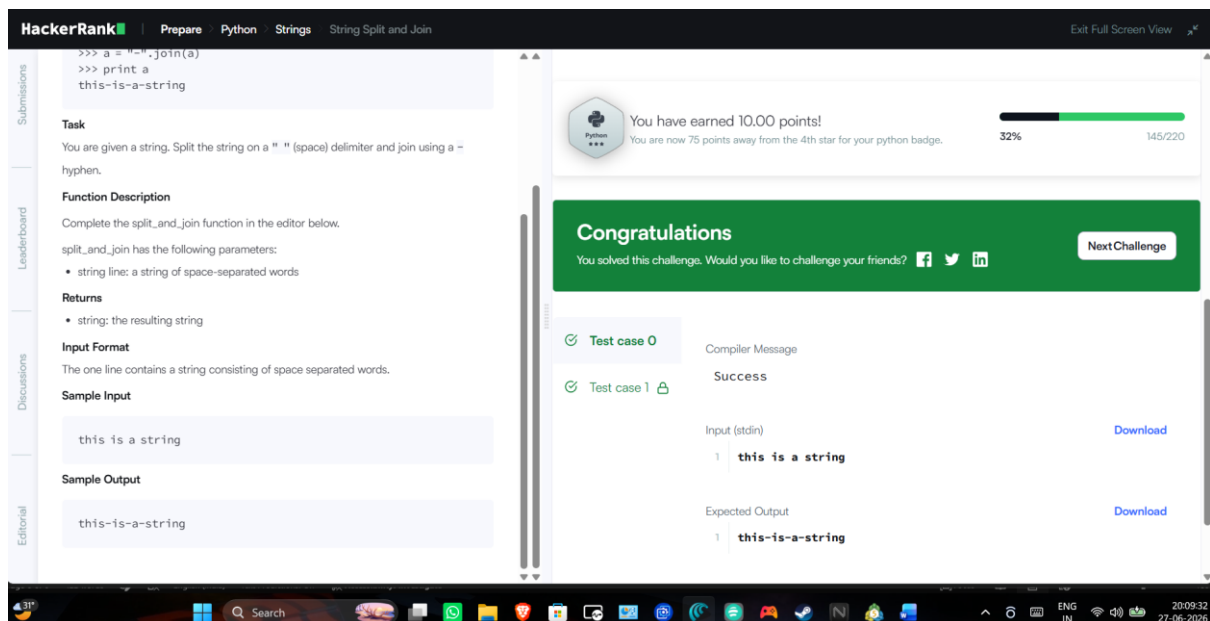
Sample Output
this-is-a-string

Change Theme | Language: Pypy 3

```
1
2
3 def split_and_join(line):
4     # write your code here
5     line = line.split(" ")
6     line = "-".join(line)
7     return line
8
9 if __name__ == '__main__':
10     line = input()
11     result = split_and_join(line)
12     print(result)
```

Line: 7 Col: 16

Upload Code as File | Test against custom input | Run Code | Submit Code



Task 4

Given an integer, n , print the following values for each integer 1 from n to :

1. Decimal
2. Octal
3. Hexadecimal (capitalized)
4. Binary

Function Description

Complete the `print_formatted` function in the editor below.

`print_formatted` has the following parameters:

- `int number`: the maximum value to print

Prints

The four values must be printed on a single line *in the order specified above* for each from to . Each value should be space-padded to match the width of the *binary* value of and the values should be separated by a single space.

Input Format

A single integer denoting .

Constraints

- $1 \leq n \leq 99$

Sample Input

17

Sample Output

1	1	1	1
2	2	2	10
3	3	3	11
4	4	4	100
5	5	5	101
6	6	6	110
7	7	7	111
8	10	8	1000
9	11	9	1001
10	12	A	1010
11	13	B	1011
12	14	C	1100
13	15	D	1101
14	16	E	1110
15	17	F	1111
16	20	10	10000
17	21	11	10001

Submissions

Leaderboard

Discussions

Editorial

HackerRank

PreparePythonStringsString Formatting

Exit Full Screen View

Constraints

- $1 \leq n \leq 99$

Sample Input

17

Sample Output

1	1	1	1
2	2	2	10
3	3	3	11
4	4	4	100
5	5	5	101
6	6	6	110
7	7	7	111
8	10	8	1000
9	11	9	1001
10	12	A	1010

Change ThemeLanguagePypy 3

```
1 def print_formatted(number):
2     # your code goes here
3     width = len(bin(number)[2:])
4     for i in range(1,n+1):
5         print(str(i).rjust(width),
6               oct(i)[2:].rjust(width),
7               hex(i)[2:].upper().rjust(width),
8               bin(i)[2:].rjust(width))
9
10
11 if __name__ == '__main__':
12     n = int(input())
13     print_formatted(n)
```

Line: 8 Col: 38

Upload Code as File

Test against custom input

Run Code

Submit Code

hackerrank.com / String Formatting

Exit Full Screen View

HackerRank

PreparePythonStringsString Formatting

Submissions

Leaderboard

Discussions

Editorial

Constraints

- $1 \leq n \leq 99$

Sample Input

17

Sample Output

1	1	1	1
2	2	2	10
3	3	3	11
4	4	4	100
5	5	5	101
6	6	6	110
7	7	7	111
8	10	8	1000
9	11	9	1001
10	12	A	1010

You have earned 10.00 points!

You are now 65 points away from the 4th star for your python badge.

41%

155/220

Congratulations

You solved this challenge. Would you like to challenge your friends?

[f](#)[t](#)[in](#)

Next Challenge

Test case 0

Compiler Message

Success

Test case 1

Test case 2

Test case 3

Input (stdin)

Download

Download

Expected Output

Download